Z: The sequence number of the course offered in same subject category

For Example, Subject having course code "MIC101" represent

1: Course offered in First year

0: Course subject category is Fundamental courses

1: The sequence number of the course offered in Fundamental courses category

MIC101 – Fundamentals of Microbiology 3(2+1)

Department of Microbiology

Kohat University of Science & Technology, Kohat

Course Title	Fundamentals of Microbiology
Course Code	MIC101
Credit Hours	3(2+1)
Contact Hours	4 Hours per Week (2 Hours Theory + 2 Hours Practical)
Semester Duration	16 Weeks
Mid Term Examination	8th Week
Final Term Examination	16th Week
Program	BS Microbiology
Semester	1st

1	To provide basic understanding of Microbiology.
2	To compare and distinguish the basic groups of microbes, including bacteria, fungi, protozoans and viruses.
3	To study the detailed structure of microbes.
4	To describe the tools and methods that is used to classify and identify microbes.

1	Have an overview about the role of microorganisms in nature.
2	Know the basic techniques used in microbiology

Week	Lecture/Practical	Course Contents
1st	Lecture - 1	Introduction to Microbiology

	Lecture - 2	History and scope of Microbiology
	Practical - 1	Basic laboratory safety guidelines
2nd	Lecture - 3	Importance of microbes and Microbial diversity
	Lecture - 4	Branches of microbiology
	Practical - 2	Equipment's used in Microbiology lab
3rd	Lecture - 5	Difference between prokaryotes and eukaryotes
	Lecture - 6	Detailed structure of bacterial cell
	Practical - 3	Use of microscope and its care
4th	Lecture - 7	Detailed structure of bacterial cell
	Lecture - 8	Archaeabacteria
	Practical - 4	Gram staining
5th	Lecture - 9	Bacterial nutrition
	Lecture - 10	Growth requirements of bacteria
	Practical - 5	Acid fast staining
6th	Lecture - 11	Types of media for bacterial growth
	Lecture - 12	Methods of bacterial culture and identification
	Practical - 6	Sterilization of equipment
7th	Lecture - 13	Introduction to viruses
	Lecture - 14	Structure and replication of viruses
	Practical - 7	Preparation of media for bacteria
8th	**Mid Term Examination**	
9th	Lecture - 15	Introduction to fungi and their importance
	Lecture - 16	Detailed structure of fungi
	Practical - 8	Isolation of bacteria from different samples
10th	Lecture - 17	Methods of culture and identification of fungi
	Lecture - 18	Methods of culture and identification of fungi
	Practical - 9	Isolation and identification of bacteria
11th	Lecture - 19	Introduction to protozoa and their importance
	Lecture - 20	Basic structure of protozoa
	Practical - 10	Biochemical tests
12th	Lecture - 21	Introduction to algae and importance
	Lecture - 22	Prions
	Practical - 11	Biochemical tests
13th	Lecture - 23	Physical methods of control of microorganisms
	Lecture - 24	Chemical methods for control of microorganisms

	Practical - 12	Preservation of bacterial cultures
14th	Lecture - 25	Antimicrobials, antifungals and antiviral agents
	Lecture - 26	Different classes of antibiotics
	Practical - 13	Isolation and identification of fungi
15th	Lecture - 27	Mechanism of actions of different antibiotics
	Lecture - 28	Antimicrobial resistance
	Practical - 14	Physical and chemical methods for control of microbes
16th	**Final Term Examination/Result Submission**	

Recommended Books:

1. Prescott's Microbiology, 10th Edition. 2012. By Joanne Willey and Linda Sherwood and Christopher J. Woolverton ISBN10:1259281590.
2. Pelczar. 5th Edition. 2010. Microbiology: Application Based Approach. Tata McGraw-Hill Education. Pp 897.
3. David White, James Thomas Drummond, Clay Fuqua. 2011. The Physiology and Biochemistry of Prokaryotes. Oxford University Press, Incorporated. 4th Ed. Pp 632.
4. Albert G. Moat, John W. Foster, Michael P. Spector. 2003. Microbial Physiology. John Wiley and Sons. Pp 736
5. Jens H. Petersen. 2013. The Kingdom of Fungi. Princeton University Press. Pp 272.